

# Timeline Check In

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## Setup

## Packages

## Data

## Ideas for analysis and background

Our project focuses on salt marsh vegetation at North Campus Open Space (NCOS), with the goal of understanding which plant species characterize this restored wetland habitat and how those patterns have changed across survey years. Rather than treating vegetation cover as only a set of numbers, we are using percent cover to describe how the salt marsh is developing as a plant community over time. In this sense, our project asks how restoration becomes visible through the species that take root, persist, expand, or decline across the marsh.

Salt marsh vegetation is especially useful for understanding restoration trajectory because these plant communities are shaped by strong environmental filters, including salinity, tidal flooding, saturated soils, and seasonal changes in water availability. In southern California salt marshes, native halophytes such as pickleweed (*Salicornia pacifica*) and saltgrass (*Distichlis spicata*) often help define the structure and identity of the habitat (Ferren 1985; North Campus Open Space 2025). Because these species are adapted to stressful coastal wetland conditions, changes in their mean percent cover can help indicate whether the restored salt marsh is developing toward a recognizable native marsh community.

This question is particularly relevant at NCOS because the site was restored from the former Ocean Meadows Golf Course into a mosaic of wetland and upland habitats, including salt marsh, brackish marsh, seasonal wetlands, and transitional habitats (Cheadle Center for Biodiversity and Ecological Restoration and Environmental Science Associates 2016; Rickard 2024). Broader NCOS monitoring reports evaluate restoration progress across habitat types, while our project narrows in on salt marsh vegetation specifically. By focusing on dominant

species and native/non-native cover through time, we can connect the survey data to larger questions about restoration progress, habitat identity, and community development.

Previous work on California wetland restoration has shown that vegetation establishment can be slow, variable, and strongly shaped by local site conditions such as elevation, soil salinity, soil moisture, planting, and adaptive management (Beheshti et al. 2023). This matters for our analysis because shifts in dominant plant cover are not just visual patterns in a graph. They may reflect ecological processes such as native plant establishment, non-native plant expansion, hydrologic change, disturbance, or interannual variability in rainfall and growing conditions.

## Ecological background

## Planned analyses

## Data cleaning and wrangling

## Summaries for planned analyses

## Visualizations directly relevant to answering questions

Figure 1: Dominant species percent cover across survey years

```
fig1 <- ggplot(data = dominant_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = psoc_grouped)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c(
    "Salicornia_pacifica"      = "#0D5E45",
    "Distichlis_spicata"      = "#1D9E75",
    "Atriplex_lentiformis"    = "#5DCAA5",
    "Eleocharis_macrostachya" = "#A8DBC9",
    "Raphanus_sativus"       = "#D85A30"
  )) +
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Species") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"),
        legend.title = element_text(face = "bold"))
```

fig1

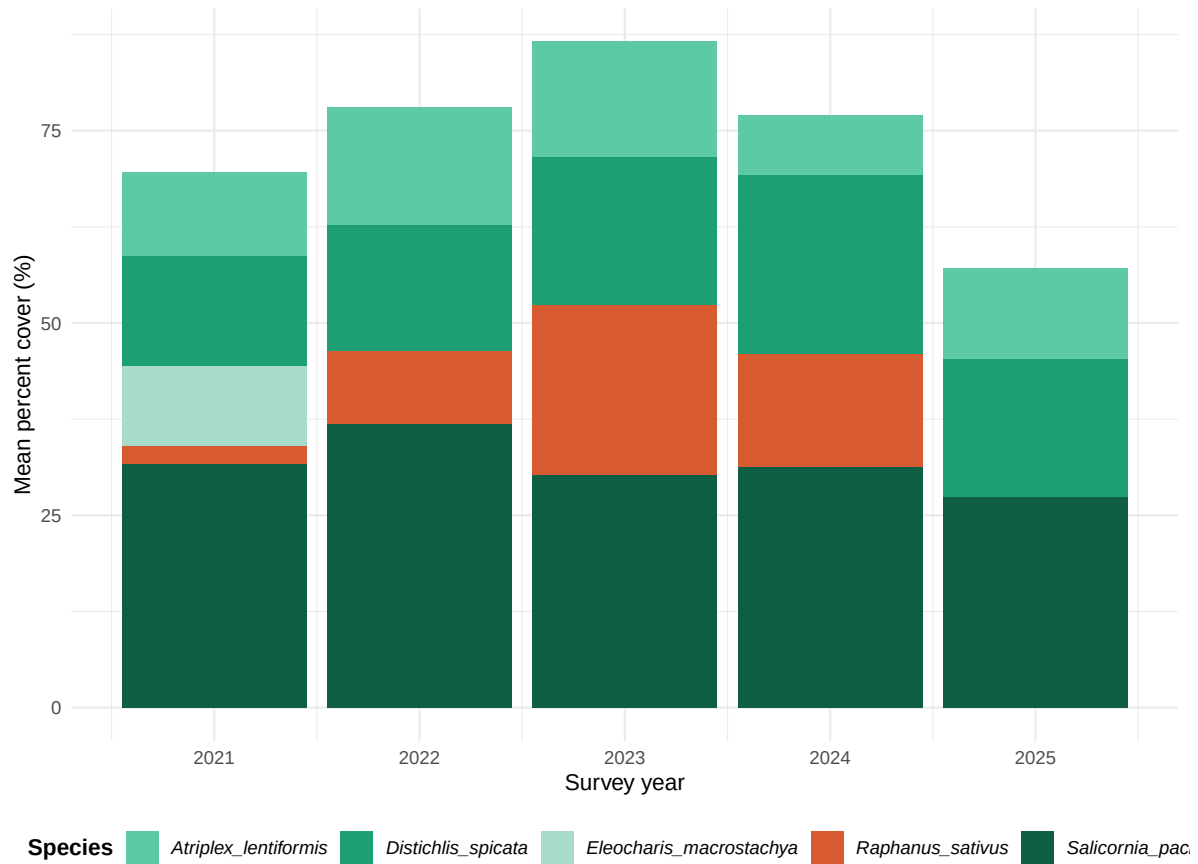


Figure 1: Mean percent cover of the five dominant salt marsh plant species at NCOS across survey years, averaged across transects within each year. Dominant species were identified as the top five by mean percent cover across all survey years. Bar height reflects total mean cover of the five dominant species and varies across years. Green shades indicate native species; coral indicates non-native species.

Figure 2: Proportional native vs. non-native cover across survey years

```
fig2 <- ggplot(data = native_nonnative,
               aes(x = year,
                   y = prop,
                   fill = cover_category)) +
```

```
geom_col(position = "stack") +
scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                             "NON-NATIVE COVER" = "#D85A30")) +
scale_y_continuous(labels = scales::percent) +
labs(x = "Survey year",
     y = "Proportion of vegetation cover",
     fill = "Cover category") +
theme_minimal() +
theme(legend.position = "bottom")
```

fig2

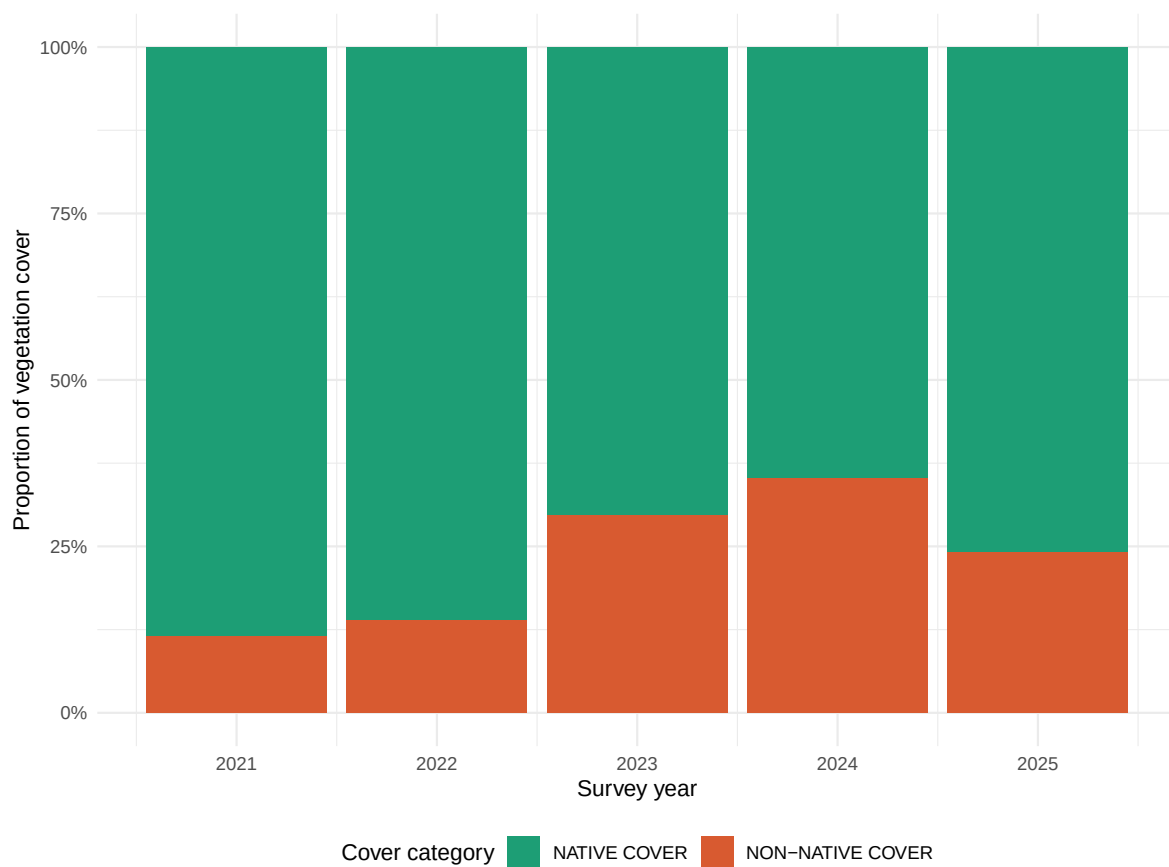


Figure 2: Proportional contribution of native and non-native vegetation cover in salt marsh habitat at NCOS across survey years. Each bar sums to 1. Native cover is shown in green; non-native cover in coral.

**Figure 3: Species rank plot**

```
# fig3 - pointrange instead of col
cutoff_x <- mean(c(species_rank$mean_pc[5], species_rank$mean_pc[6]))

fig3 <- ggplot(data = species_rank_top20,
               aes(x = mean_pc,
                   y = reorder(psoc, mean_pc),
                   color = cover_category)) +
  geom_pointrange(aes(xmin = mean_pc - se_pc,
                      xmax = mean_pc + se_pc)) +
  geom_vline(xintercept = cutoff_x,
             linetype = "dashed",
             color = "gray50") +
  scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                                "NON-NATIVE COVER" = "#D85A30")) +
  labs(x = "Mean percent cover ( $\pm$  1 SE)",
       y = "Species",
       color = "Cover category") +
  theme_minimal() +
  theme(axis.text.y = element_text(face = "italic", size = 9),
        legend.position = "bottom",
        legend.title = element_text(face = "bold"))

fig3
```

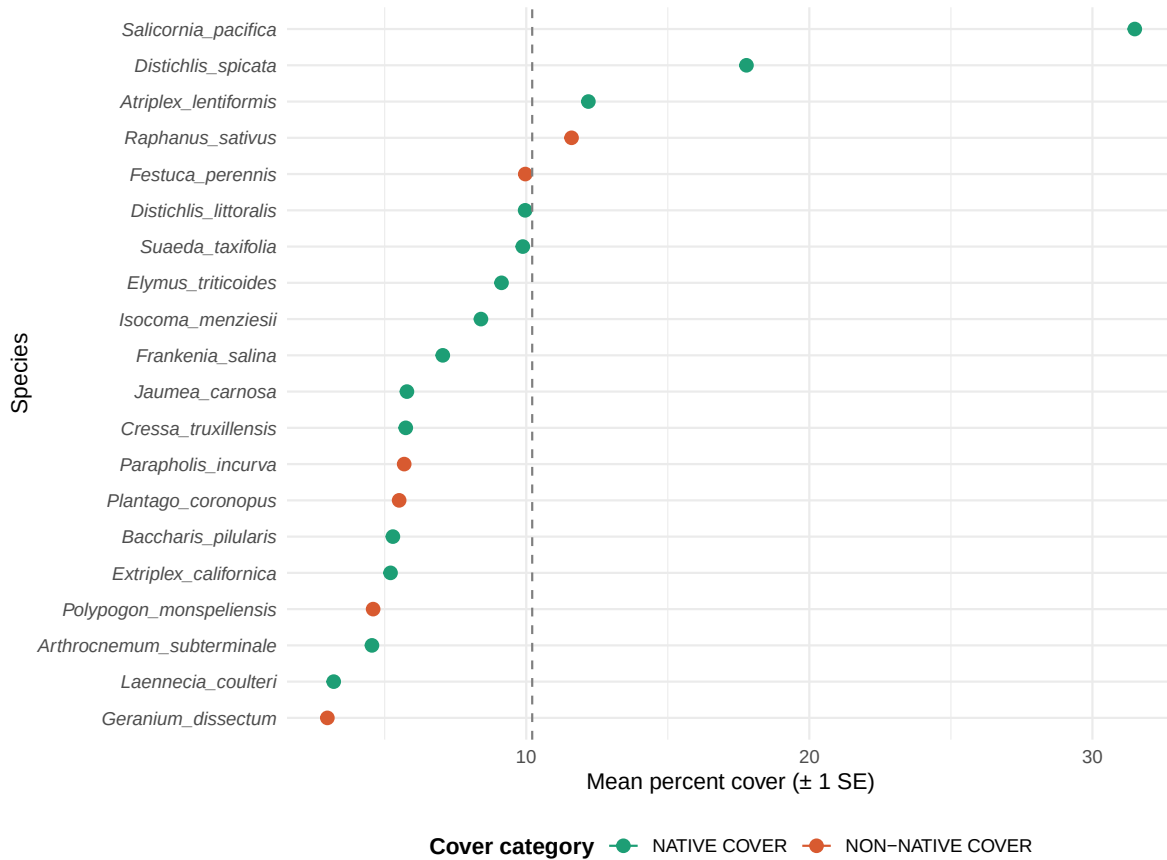


Figure 3: Mean percent cover ( $\pm 1$  SE) of the top 20 plant species recorded in salt marsh habitat at NCOS, averaged across transects and survey years and ranked from highest to lowest mean cover. Points represent means and horizontal bars represent  $\pm 1$  standard error. Green indicates native species; coral indicates non-native species. Dashed line marks the cutoff for the five dominant species visualized in figure 1.

## Other exploratory visualizations

Figure 4: Total vegetation cover across survey years

```
fig4 <- ggplot(data = total_cover,
               aes(x = year,
                   y = mean_total)) +
  geom_ribbon(aes(ymin = mean_total - se_total,
                  ymax = mean_total + se_total),
```

```

    fill = "#1D9E75",
    alpha = 0.2) +
geom_line(color = "#1D9E75",
    linewidth = 1) +
geom_point(color = "#1D9E75",
    size = 3) +
labs(x = "Survey year",
    y = "Total mean percent cover (%)") +
theme_minimal()

```

fig4



Figure 4: Total mean percent vegetation cover (native + non-native combined) in salt marsh habitat at NCOS across survey years. Shaded band represents  $\pm 1$  standard error around the mean.

## Plan for elective (if completing)

Elective concept

Draft sections (include some description/image of drafts)

Plan for the next two weeks

## Next steps

## References

Beheshti, Kathryn M., Stephen C. Schroeter, Andres A. Deza, Daniel C. Reed, Rachel S. Smith, and Henry M. Page. 2023. "Large-Scale Field Studies Inform Adaptive Management of California Wetland Restoration." *Restoration Ecology* 31 (5): e13936. <https://doi.org/10.1111/rec.13936>.

Cheadle Center for Biodiversity and Ecological Restoration, and Environmental Science Associates. 2016. *North Campus Open Space Restoration Project Restoration Plan*. University of California, Santa Barbara. <https://escholarship.org/uc/item/9505h60h>.

Ferren, Wayne R. Jr. 1985. *Carpinteria Salt Marsh: Environment, History, and Botanical Resources of a Southern California Estuary*. Publication Number 4. The Herbarium, Department of Biological Sciences, University of California, Santa Barbara.

North Campus Open Space. 2025. *Coastal Salt Marsh*. <https://www.ncos.ccber.ucsb.edu/native-plant-habitats/coastal-salt-marsh>.

Rickard, Alison. 2024. *North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024*. University of California, Santa Barbara Cheadle Center for Biodiversity; Ecological Restoration. <https://escholarship.org/uc/item/7bq618m8>.